

FormSpace — One-Page Reference

The mathematical object that unifies the four ITDX26 tasks

D1 · Latent physical state

Sensor observations y_t^m from M modalities are noisy views of a partially observed dynamical system with hidden state s_t :

$$y_t^m = h_m(s_t, c_t, \kappa_t^m) + \varepsilon_t^m$$

D2 · The signal map Λ

NLM learns a modality-agnostic map $\Lambda: X \rightarrow Z$ from calibrated multi-modal signals to a latent embedding, trained with masked reconstruction, contrastive alignment, physics regularization, and calibration losses.

D3 · Form Space

Two observations are Form-equivalent when their embeddings agree under the learned tolerance:

$$x_1 \sim_\Lambda x_2 \iff \Lambda(x_1) \approx \Lambda(x_2); F_\Lambda = X / \sim_\Lambda$$

Form Space is the space of physically distinguishable states the model resolves at its current data budget. Resolution improves with data; the object is learned, not metaphysical.

D4 · Admissibility gate

A deterministic function decides which points in Form Space are permitted operating regions:

$$G(E, \phi, v, A, R) \rightarrow \{\text{PASS, REVIEW, DENY, PAUSE}\}$$

where E is evidence sufficiency, ϕ is policy, v is environmental stewardship, A is the action envelope, and R is the rollback plan. AVANI enforces; MYCA proposes; human dispositions the result.

The four ITDX tasks as operations on F_Λ

- Task 12 · Pattern — equivalence-class discovery in F_Λ .
- Task 13 · Link — typed hypergraph inference over Form-classes.
- Task 8 · COAs — constrained trajectories on the admissible submanifold.
- Task 14 · Map — deterministic projection of F_Λ onto Army coordinates.

NLM = Nature Learning Model (not Natural Language for Mycology). FormSpace is Mycosoft's signal-native modeling framework for living and physical Earth systems.